LESSON PLAN

WEAPON GUIDANCE (19)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	21
Total No. of Lesson Plans	22	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Practical on aircraft wiring routing and repair like crimping, insulating using heat sleeve, Solder sleeve etc.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Practical on aircraft wiring routing and repair like crimping, insulating using heat sleeve, Solder sleeve etc.
5. Specific Objectives	19.1.1.1Trainee will be able to explain aircraft wiring and Causes of wiring degradation.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Hands-on practice on crimping terminals, applying heat shrink sleeves, and using solder sleeves on aircraft wiring samples.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary causes of aircraft wiring degradation?
(b) Why is proper crimping essential for electrical continuity?
(c) What safety precautions must be observed during soldering operations?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Preparation of wire ends and stripping techniques • Selection and application of crimping tools • Inspection of crimp quality and terminal fit • Application of heat shrink insulation sleeves • Use of solder sleeves for additional protection	L & Ds	PPT, IB	15 min
(b)	• Routing wires through aircraft structure • Securing wires with clips and ties • Avoiding interference with moving parts • Maintaining proper bend radius • Labeling and identification of repaired wires	L & Ds	PPT, PDF	20 min
(c)	• Final inspection of repaired wiring • Testing continuity and insulation resistance • Documentation of repair details • Cleaning work area and tools • Review of common defects and corrections	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Describe the correct procedure for applying a heat shrink sleeve to ensure a proper seal.
(b) Identify three signs of a defective crimp and explain how to correct each.
(c) Explain why solder sleeves are used in addition to heat shrink insulation in certain applications.

12. Home Assignment and Follow Up	Prepare a brief report detailing the steps for routing a wire bundle in a confined space, referencing the procedures to be covered in LP 22 regarding connector pin insertion.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Practical on aircraft connectors like pin configuration, insertion and exertion of connector pins, insulation check.	Refer to the master syllabus reference list	LP 22

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min